Problem Set 1

Erica Lundt

October 8, 2026

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
    80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Answers For Question 1.1

First, determining length which is the number of values that we have in this data set.

```
length(y)
```

ANSWER FOR THE LENGTH OF Y IS: 25

The second is to determine the mean of sample the scores.

```
mean_score <- mean(y)

mean_score
```

ANSWER OF MEAN SCORE FOR THE SCORES:98.44

The third is to determine the standard deviation of the scores.

```r
sd_score <- sd(y)

sd_score
```

ANSWER OF SD SCORE FOR THE SCORES: 13.09287

I have calculated the length, mean and SD. In order to determine the confidence interval, the standard error will also have to be calculated.

```r
se_score <- sd_score / sqrt(length(y))

se_score
```

ANSWER FOR THE STANDARD ERROR SCORES: 2.618575

Now having together the length, mean, standard deviation, and the standard error we can now calculate the confidence interval at 90%

CI = mean +/- critical value * SE

qnorm gives the z-value given proportion of the standard normal distribution

```r
qnorm(0.95)
```

ANSWER TO THE UPPER QNORM IS: 1.644854

```r
qnorm(0.05)
```

ANSWER TO THE LOWER QNORM IS: -1.644854

Calculating the CI

```r
lower_90 <- qnorm(0.05,
                  mean = mean_score,
                  sd = se_score)
```

ANSWER FOR THE LOWER TAIL: 94.13283

```r
upper_90 <- qnorm(0.95,
                  mean = mean_score,
                  sd = se_score)
```

ANSWER FOR THE UPPER TAIL: 102.7472

Final Answer for Question 1.1: At the CI level of 90 percent, the upper score average is approximately 103. While the lower average of the IQ score is 94.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.
Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Answers For Question 1.2

Hypothesis testing will begin with two statements:

Null hypothesis is: the average student's IQ is higher than 100 among all

schools in the country.

HO greater 100

Alternative hypothesis is: the average student's IQ is not higher than 100 among all schools in the country.

HA less than 100

This will be at one-sided test because we are looking at if something is or is not the value of the null.

The test statistic is calculated to determine the how far the estimate is from the null hypothesis.

Test = estimate - null value / SE

T-statistic will be used due to the small population size (i.e 25 data points) and because this is a one-sided test.

The test will look at y (data sampling) and the degrees of freedom. With this we can observe if the we can reject the null hypothesis of HO greater than 100 through the p-value outcome.

```
t.test(y, mu = 100,alternative = "greater") #H0 ~ p-value of 0.7215

t.test(y, mu = 100,alternative = "less") #HA ~ p-value of 0.2785
```

Final Answer for 1.2: We failed to reject the null at the standard significance rate of 0.05 because the p-value is greater than the standard used. Thus, we cannot conclude the average IQ of the counselor's school is greater than the country's IQ scores.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

```
head(expenditure)
```

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

- # Answers For Question 2.1

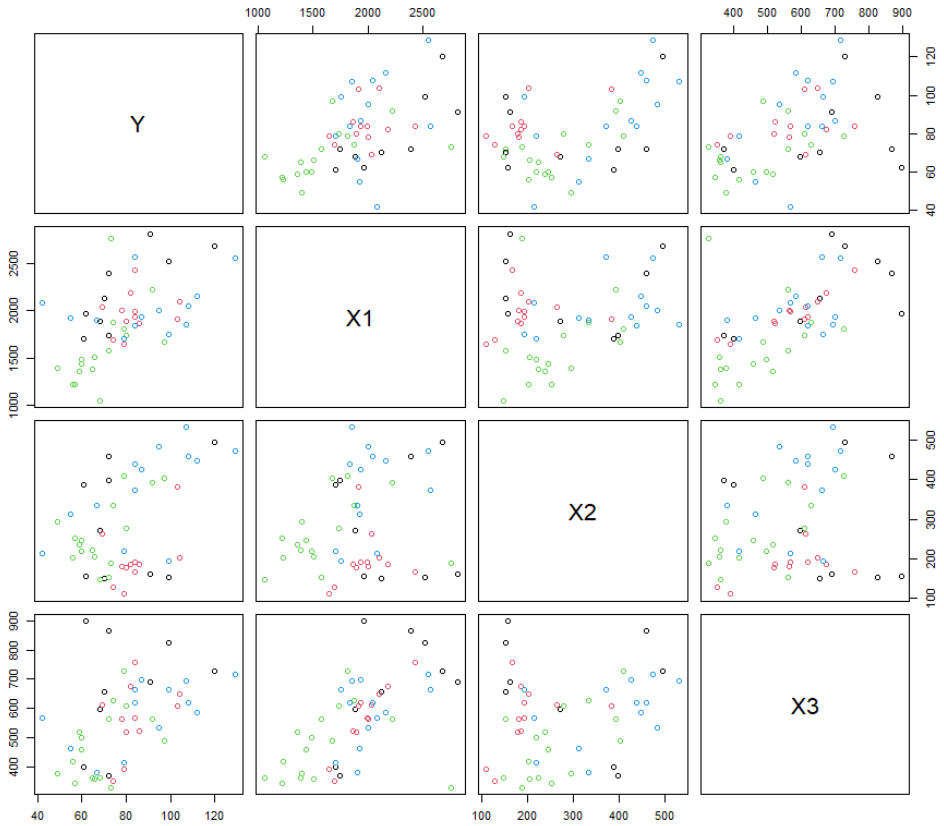


Figure 1: Housing Assistance and Income, Finance, and Region.

Assessment of all Plots:

```
#Taking all of these into account of all three plots, there is slight
    correlation that increases in capita expenditure can increase well
    being in Urban areas.
```

Answers For Question 2.2

Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```
# On average the region that has the most shelter/housing assistance is
    region 4, the West.
```

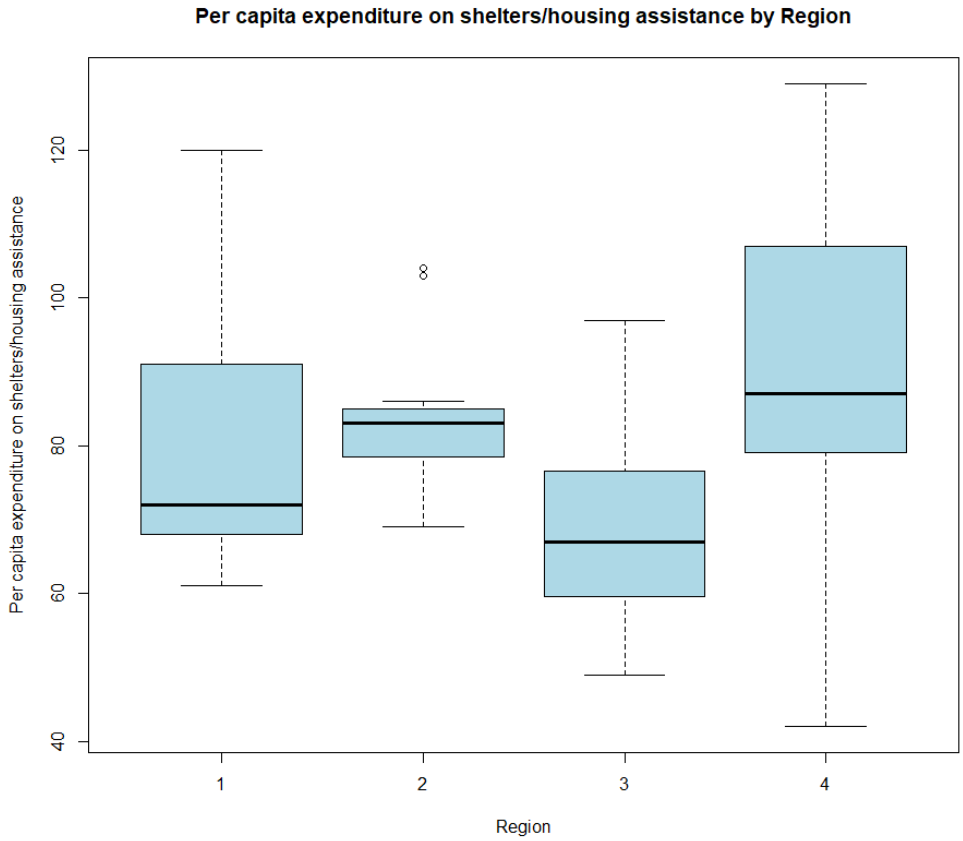


Figure 2: Housing Assistance and Region.

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

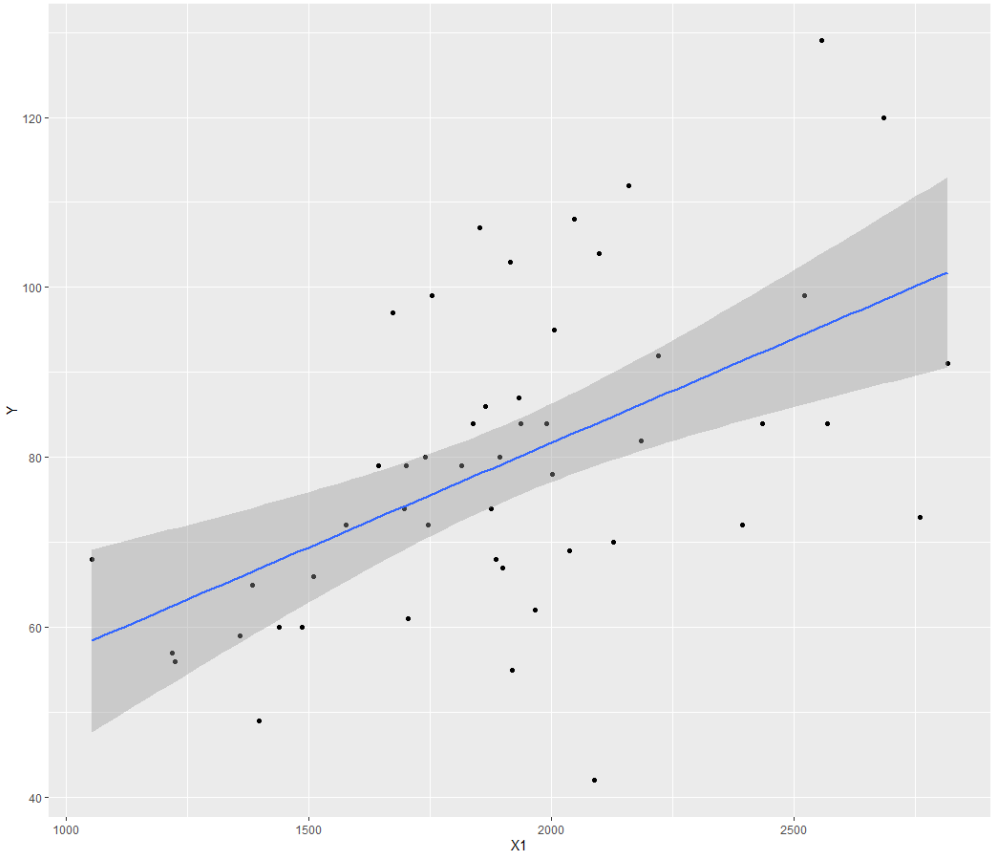


Figure 3: Housing Assistance and Region

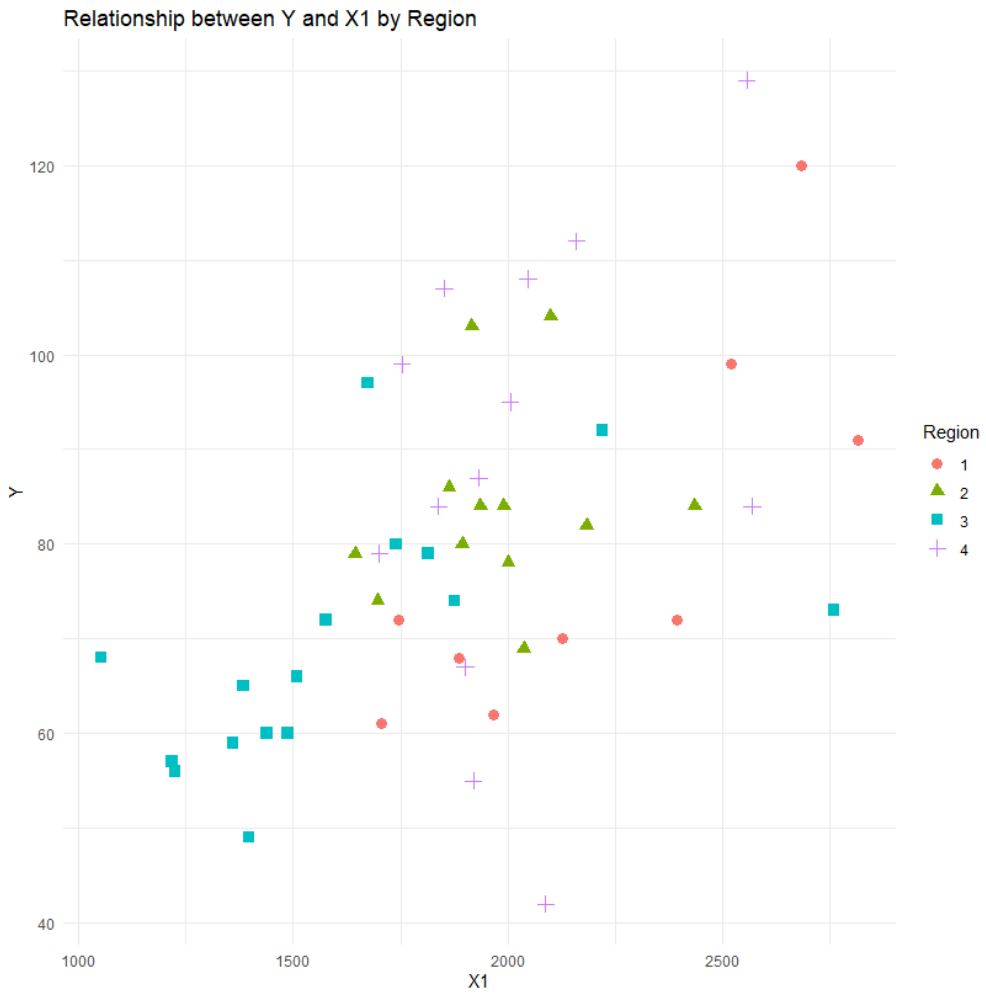


Figure 4: Housing Assistance and Region Part 2.

```
# The relationship between Y and X1 shows that the more capital put into
    shelter and housing assistance, the income stays around $2000. There is
    steady growth. When adding  an additional variable of regions you see that
     region 3 (the south), puts in low capita for housing and shelter and have
     lower income.
```